

Problems

INTERNATIONAL MATHEMATICS SUMMER CAMP IMSC 2025 MOCK TEST 2-GEOMETRY



Date: Monday, 30th June 2025
Number of problems: 3

Time: 13:30-18:00
Total points: 21

Problems

Problem 1. In an isosceles triangle ABC with $AB = AC$, the angle bisector of angle B intersects side AC at point B' , and the equality $BB' + B'A = BC$ holds. Determine the measures of the angles of triangle ABC .

Problem 2. Suppose ABC be a triangle and let A', B', C' be the feet of the altitudes on the sides BC, CA and AB , respectively. Let H be the orthocenter and let M be the midpoint of AH . Points X and Y are the feet of perpendiculars from A' to MB' and MC' , respectively. Prove that B, C, X, Y lie on the same circle.

Problem 3. Let ABC be a triangle and consider the A, B, C -mixtilinear circles centered at O_1, O_2, O_3 and tangent to the circle ABC at points D, E, F , respectively. Lines FO_2 and EO_3 meet at point X , lines DO_3 and FO_1 meet at point Y , and lines EO_1 and DO_2 meet at point Z . Prove that lines XO_1, YO_2 and ZO_3 are concurrent.

Note: The A -mixtilinear circle is the circle tangent to the segments $(AB), (AC)$ and internally tangent to the circumcircle $\odot(ABC)$.

Solution 1. On side BC of the triangle, consider the point M such that $BB' = BM$. The given equality becomes $AB' = MC$.

The angle bisector theorem implies

$$\frac{AB'}{B'C} = \frac{AB}{BC} = \frac{AC}{BC},$$

from which

$$\frac{MC}{B'C} = \frac{AC}{BC}.$$

It follows that triangles MCB' and ACB are similar, hence $MC = MB'$.

Furthermore, $\angle C = \angle MCB' = \angle MB'C$. Then $\angle BMB' = 2C$, and since triangle $BB'M$ is isosceles, it follows that $\angle BB'M = 2C$.

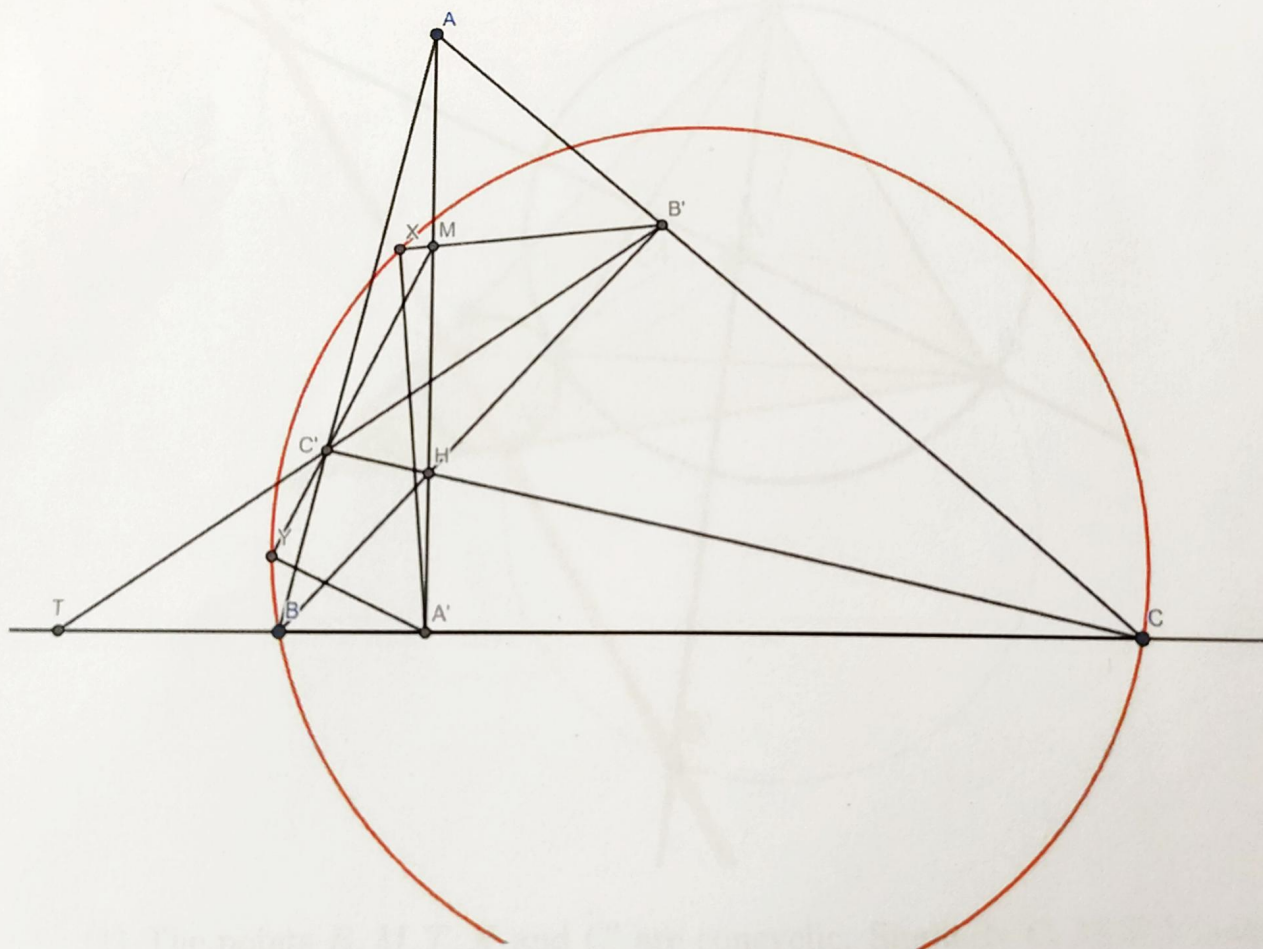
In triangle $BB'M$ we have $180^\circ = 2C + 2C + C/2$, that is

$$C = B = 40^\circ, A = 100^\circ.$$



Consider a special case when ABC is isosceles at A . Let A' be the antipode of A . Then the orthocenter H is the reflection of A' through BC and the Simson line of P wrt ABC will bisect PH . Hence if we call T the intersection of PA' and BC , then the Simson line of P wrt ABC will bisect PT .

Since $(T, A'; B, C) = -1$, it follows that $NA'^2 = NB \cdot NC$. On the other hand, since the center of the circumcircle of triangle $A'XY$ is the midpoint of MA' , that circle has to touch BC at A' . Thus, $NA'^2 = NX \cdot NY$. Therefore, $NX \cdot NY = NB \cdot NC$, or equivalently, B, C, X, Y are concyclic.

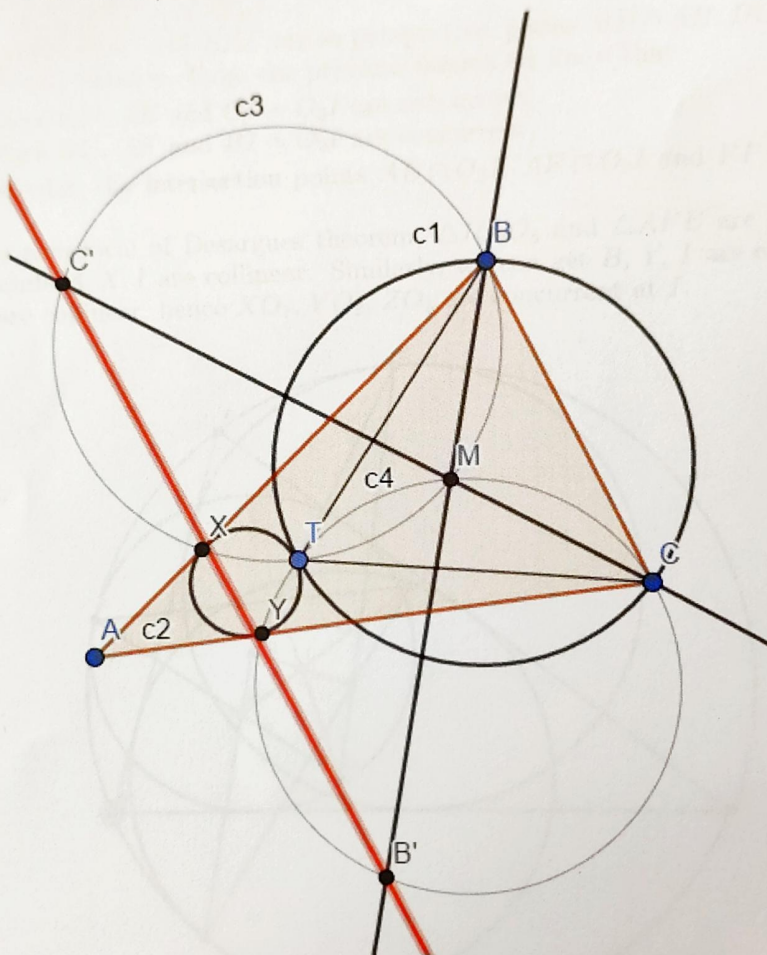


Solution 3.

Some properties on Protasov configuration will be used.

Protasov's Lemma: Let ABC be a triangle with incenter I . Let c_1 be a circle passing through B, C and intersecting the segments $[AB]$ and $[AC]$ again. Let c_2 be the circle tangent to $(AB), (AC)$ and externally tangent to c_1 . Denote T the tangency point of c_1 with c_2 . Then, TI is the angle bisector of $\angle BTC$.

Moreover, let X, Y be the tangency points of c_2 with AB and AC respectively; and let $B' = BI \cap XY, C' = CI \cap XY$. Then we have the following properties:



- (1) The points B, M, T, X and C' are concyclic. Similarly C, M, T, Y and B' .
- (2) The points B, C, B' and C' are concyclic.

Let I be the incenter of $\triangle ABC$. Let $X_1 = O_2O_3 \cap BC$, $X_2 = O_3O_1 \cap CA$, $X_3 = O_1O_2 \cap AB$. Let $P = BF \cap CE$, $Q = AE \cap BD$, $R = CD \cap AF$. Let M, N, M', N' be the midpoint of arcs AC, AB, ABC and ACB .

By D'Alembert theorem we get X_1, X_2, X_3 are collinear. Let $\ell = X_1X_2$. By Pascal theorem (for $EMN'FNM'$) we get $A, I, ME \cap NF$ are collinear. By Pascal

theorem (for $BFNCEM$) we get A, I, P are collinear, i.e. P lie on the bisector of $\angle BAC$. By Pascal theorem (for $FBACED$) we get P lie on ℓ . Similarly, we can get Q lie on CI and ℓ , R lie on BI and ℓ .

Lemma. Lines AE , BD and CI are concurrent.

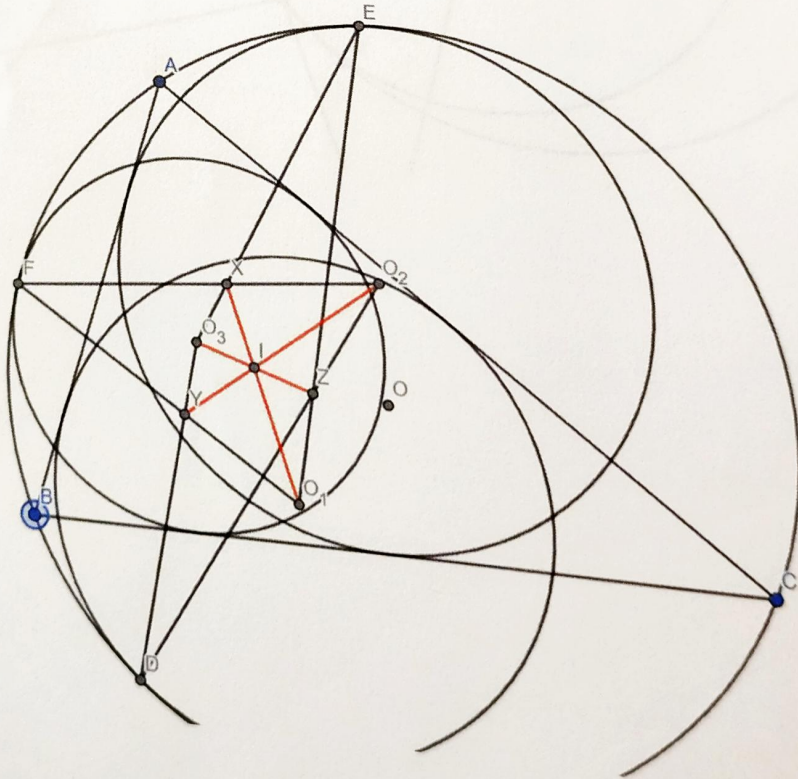
Proof of the lemma. Let the A -mixtilinear incircle touch AB, AC at A_C, A_B . It's well-known that I is the midpoint of $A_B A_C$. In particular $I = CI \cap A_B A_C$. It is then a consequence of the Protasov Configuration that CI is tangent to (IBD) . Similarly CI is tangent to $(IAE) \Rightarrow AE, BD, CI$ concur at the radical center of circles $(ABC), (IBT_A), (IAT_B)$. $\square$

As triangles BDC and EAF are in perspective, points $BD \cap AE$, $DC \cap AF$ and $BC \cap EF$ are collinear. From the previous lemma we know that

- (1) Lines BD , AE and $CI = O_3 I$ are concurrent,
- (2) Lines DC , AF and $BI = O_2 I$ are concurrent.

In particular, the intersection points $AE \cap O_3 I$, $AF \cap O_2 I$ and $EF \cap O_2 O_3$ are collinear.

By the reciprocal of Desargues theorem, $\triangle IO_2 O_3$ and $\triangle AFE$ are perspective. Hence points A, X, I are collinear. Similarly, we can get B, Y, I are collinear and C, Z, I are collinear, hence XO_1, YO_2, ZO_3 are concurrent at I .



GEOMETRY

